
Adaptive Saccadic Reservoir Imaging Networks at the Edge of Chaos

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Abstract

1 Reservoir computing promises simple recurrent models with strong theory and
2 hardware appeal, yet struggles on images because static inputs must be serialized
3 via hand-designed scans that discard spatial structure and create long, redundant
4 sequences. We address this interface with Adaptive Saccadic Reservoir Imag-
5 ing Networks (ASRIN), which co-optimizes when visual information is revealed
6 and how reservoir dynamics operate. ASRIN combines a differentiable saccadic
7 encoder that selects a short sequence of foveal patches via a learned policy, an
8 edge-of-chaos self-tuning reservoir that stabilizes dynamics through an astrocyte-
9 like gain, and a dual-stream readout that fuses instantaneous patches with reservoir
10 state for early decisions. We connect ASRIN to its recurrent-kernel limit and derive
11 a content-dependent kernel K0ASRIN induced by learned trajectories, linking
12 finite reservoirs to kernel methods Dong et al. [2020], Nestler et al. [2020]. We
13 implement a complete protocol on MNIST-family datasets with robustness and
14 dynamical metrics and conduct a smoke-test run to validate feasibility. In this
15 undertrained configuration, ASRIN executes learned scan trajectories and reduces
16 compute by over $20\times$ versus raster reservoirs, while accuracy remains at chance.
17 We report accuracy, calibration, robustness deltas, Lyapunov and memory capac-
18 ity estimates, and kernel-limit results, discuss limitations, and outline the steps
19 required to realize the anticipated gains Ivanov and Michmizos [2021].

20 1 Introduction

21 Reservoir computing (RC) leverages fixed recurrent dynamics with a trained linear readout to
22 process sequences at low training cost. Applying RC to static images, however, requires converting
23 two-dimensional structure into a one-dimensional stream. Common practice uses raster scans or
24 spike encoders, which discard spatial semantics, generate very long sequences that exceed fading
25 memory, and fix the reservoirs operating point to brittle regimes. This interface problem matters
26 both scientifically-active perception and criticality are central to computational neuroscience-and
27 practically, where low-power hardware benefits from short sequences and fixed weights.

28 Two obstacles persist. First, the image-to-sequence transformation should reveal task-relevant content
29 quickly and semantically; hand-crafted rasters do not. Second, reservoir performance depends on
30 its dynamical regime: near the edge of chaos, reservoirs balance memory and nonlinearity, but
31 maintaining this regime online without hyperparameter sweeps is difficult. Any solution should
32 preserve RCs simplicity (no backpropagation through recurrent weights) and remain theoretically
33 analyzable under content-dependent inputs. Prior work connects random reservoirs to recurrent
34 kernels and structured transforms Dong et al. [2020] and reformulates random recurrent dynamics
35 via temporal kernels Nestler et al. [2020]. In spiking systems, astrocyte-modulated plasticity self-
36 organizes near-critical dynamics and improves accuracy without dataset-specific tuning Ivanov
37 and Michmizos [2021]. Studies of RNN bifurcations and stability underscore the need to control

dynamical regimes Eisenmann et al. [2023], and on-sensor sequential vision demonstrates the value of selecting informative measurements early So et al. [2023]. Yet, a learned image-to-sequence interface coupled with online criticality control and a recurrent-kernel analysis of content-dependent streams has been missing.

We propose Adaptive Saccadic Reservoir Imaging Networks (ASRIN). The key idea is to jointly learn where and when to look while self-tuning the reservoir towards near-critical dynamics. A differentiable saccadic encoder selects a short sequence of foveal patches via a learned policy, preserving spatial semantics and drastically reducing sequence length. An edge-of-chaos self-tuning reservoir adds an astrocyte-like gain to maintain a sensitive regime online. A dual-stream readout combines instantaneous evidence with temporal state for early classification. Finally, we analyze the recurrent-kernel limit and derive a content-adaptive kernel K_{ASRIN} induced by the learned trajectories, linking finite reservoirs to kernel machines Dong et al. [2020], Nestler et al. [2020].

- **Learned saccadic encoder:** A differentiable, learned saccadic encoder that converts images into short, semantically structured sequences co-optimized with a reservoir readout.
- **Edge-of-chaos self-tuning:** An edge-of-chaos self-tuning mechanism that regulates a global recurrent gain online, inspired by astrocyte-modulated dynamics, without training recurrent weights Ivanov and Michmizos [2021].
- **Dual-stream readout:** A dual-stream readout that disentangles instantaneous patch evidence from temporal state and supports early decisions and ablations.
- **Recurrent-kernel analysis:** A recurrent-kernel analysis yielding K_{ASRIN} , a content-adaptive kernel induced by learned saccades, extending recurrent kernels to adaptive input streams Dong et al. [2020], Nestler et al. [2020].
- **Protocol and smoke test:** A rigorous experimental protocol covering accuracy, calibration, efficiency, robustness, and dynamics, plus a smoke-test execution that validates feasibility and reveals failure modes.

We evaluate on MNIST, Fashion-MNIST, and EMNIST-Balanced for clean accuracy, calibration, efficiency, and robustness to Gaussian noise, rotation, occlusion, and FGSM attacks. Dynamical metrics include largest Lyapunov exponent and memory capacity. We also compare finite reservoirs with their kernel-limit counterparts. To validate feasibility within strict time budgets, we executed a severely reduced smoke test that confirms learned saccades and substantial compute savings; accuracy, however, remains at chance due to undertraining, motivating the full protocol.

ASRIN complements recurrent kernels and structured reservoirs Dong et al. [2020], analytical kernel views of random recurrent dynamics Nestler et al. [2020], astrocyte-driven near-critical liquids Ivanov and Michmizos [2021], and stability-aware RNN design Eisenmann et al. [2023], Rusch and Mishra [2021], Orhan and Pitkow [2019], while aligning with the broader theme of active sensing So et al. [2023]. Beyond classification, it opens avenues to study active perception in recurrent kernels, integrate event-based sensing, and link to spiking and quantum reservoirs Bausch [2020].

2 Related Work

Recurrent kernels and structured reservoirs. Random reservoirs with fixed weights admit kernel limits that enable closed-form training and analysis, and can be accelerated via structured transforms Dong et al. [2020]. These kernels typically assume externally defined input streams. ASRIN extends this line by learning the input stream itself through a saccadic encoder and deriving a content-dependent recurrent kernel K_{ASRIN} that conditions on the learned trajectories, thus adapting the kernel to the task-driven scan order. Complementary analytical reformulations using Greens functions reveal how stimulus statistics interact with nonlinear dynamics to shape effective temporal kernels Nestler et al. [2020].

Liquid and near-critical reservoirs. Astrocyte-modulated plasticity can self-organize spiking liquids toward the edge of chaos, improving accuracy without heavy hyperparameter tuning Ivanov and Michmizos [2021]. ASRIN adopts this principle but in a rate-based ESN with frozen synapses, introducing a scalar, gradient-free gain that regulates recurrent strength online. This preserves RC simplicity while steering dynamics, differing from approaches that train recurrent connectivity or rely on spiking substrates.

Analyses of recurrent dynamics and stability. Bifurcation analyses expose dynamical regime changes that can impede training in RNNs Eisenmann et al. [2023]. Architectures with structure-preserving discretizations can mitigate vanishing or exploding gradients Rusch and Mishra [2021], and non-normal connectivity can enhance signal propagation and memory in nonlinear RNNs Orhan and Pitkow [2019]. ASRIN does not update recurrent weights; instead, it modulates the effective connectivity online to maintain a favorable regime, complementing these architectural and training strategies.

Active sensing and on-sensor processing. PixelRNN shows that prioritizing salient spatiotemporal features at the sensor can reduce bandwidth while maintaining accuracy So et al. [2023]. ASRIN similarly prioritizes where to look over time but targets reservoir computing: learned saccades produce short sequences compatible with fading memory, and the design remains analyzable via its recurrent-kernel limit.

Robustness in spiking and recurrent models. Stability-oriented training improves robustness in spiking networks Ding et al. [2024]. ASRIN seeks robustness via two complementary means: content-adaptive saccades that expose discriminative structure early and near-critical dynamics that maintain sensitivity without instability. This differs from adversarial training and formal stability constraints but aims at similar resilience goals.

In comparison to raster or Poisson encodings used in reservoir imaging pipelines, ASRIN learns the input stream and regulates the operating point online. Relative to astrocyte-modulated liquids, it keeps synapses fixed and controls a global gain. Relative to recurrent kernels, it makes the input stream content-dependent and learned, and provides a kernel-limit analysis conditioned on these learned trajectories Dong et al. [2020], Nestler et al. [2020].

3 Background

3.1 Problem Setting

We consider image classification with reservoir computing. An image is mapped to a short sequence of foveal patches $x_1, \dots, x_T$ that drive a reservoir state $h_1, \dots, h_T \in \mathbb{R}^N$. A linear readout uses either the final or intermediate state (and, in our design, the instantaneous input) to classify. Rasterization yields T equal to the number of pixels and ignores two-dimensional structure, which dilutes the reservoirs fading memory and hinders robustness.

3.2 Echo-State Networks and Kernel Connections

Echo-state networks (ESNs) fix recurrent weights and train only a linear readout. Performance depends on spectral radius, leak, nonlinearity, and input scaling. In the large-width limit, random reservoirs admit a recurrent-kernel description, linking ESNs to kernel methods and enabling structured transforms for efficiency Dong et al. [2020]. Analytical viewpoints based on temporal kernels clarify how input statistics interact with nonlinear dynamics Nestler et al. [2020]. These connections motivate designing an image-to-sequence interface whose statistics are learned for the task and amenable to kernel analysis.

3.3 Criticality and Robustness

Near the edge of chaos, recurrent networks balance memory and nonlinearity. In spiking liquids, astrocyte-like modulation can self-tune dynamics toward criticality and improve accuracy without task-specific tuning Ivanov and Michmizos [2021]. Diagnostics include the largest Lyapunov exponent (to locate the operating regime) and memory capacity (to quantify temporal integration). Bifurcations during training can abruptly alter dynamics and impede optimization Eisenmann et al. [2023]. These insights motivate an online regulator that maintains a stable, sensitive regime while keeping training simple.

3.4 Active Perception

Selecting a few informative patches over time can compress the input while preserving semantics. On-sensor architectures show the benefits of early, structured measurements under resource constraints

138 So et al. [2023]. For reservoir imaging, a learned saccadic encoder can drastically shorten T and
 139 better align the input stream with reservoir memory.

140 3.5 Formalism

141 At each time t , a policy selects a location ct on an 8×8 grid over normalized image coordinates
 142 and extracts a $p \times p$ patch xt via bilinear sampling. The reservoir update uses a leaky tanh rule: a
 143 pre-activation combines input and recurrent drives; the candidate state applies tanh; the leak forms ht .
 144 A scalar modulatory gain gt rescales the recurrent matrix online to regulate sensitivity and approach
 145 near-criticality. A linear readout operates on the dual-stream vector $[xt; ht]$. Only the policy and
 146 readout are trained; reservoir connections remain fixed. In the recurrent-kernel limit, the random
 147 reservoir induces a Gram matrix K_ASRIN over samples that depends on the sequences $x1 : T$
 148 generated by the learned policy, enabling kernel ridge classification Dong et al. [2020], Nestler et al.
 149 [2020].

150 4 Method

151 4.1 Differentiable Saccadic Encoder

152 We define an 8×8 grid of candidate fixation centers on normalized image coordinates. At time
 153 t , a policy receives a compact context consisting of an embedding of the previous patch and a
 154 low-dimensional projection of the detached previous reservoir state. The policy is a small GRU
 155 followed by an MLP head that emits logits over 64 grid cells. Centers are sampled via straight-through
 156 Gumbel-Softmax with temperature annealed from 1.0 to 0.5, yielding discrete, stable saccades. Given
 157 a sampled center ct , a $p \times p$ foveal patch xt is extracted via bilinear sampling. The sequence length is
 158 short ($T \approx 30$), preserving spatial semantics while drastically reducing compute relative to pixel-wise
 159 rasterization.

160 4.2 Edge-of-Chaos Self-Tuning Reservoir

161 The reservoir is a sparse leaky tanh ESN with fixed input matrix W_in and fixed recurrent matrix
 162 W_rec scaled to a nominal spectral radius near 0.9. Given state $ht - 1$ and patch xt , the pre-
 163 activation is computed as $ut = W_inxt + (1 + gt) W_recht - 1$. The candidate state applies tanh,
 164 and the leaky update uses $ht = (1 - \alpha)ht - 1 + \alpha \tanh(ut)$ with leak $\alpha \approx 0.3$. We monitor an
 165 activity proxy st equal to the mean of $1 - \tanh(ut)^2$ across units. The gain updates online via a
 166 smoothed negative feedback: $gt + 1$ is obtained by adding $\eta(s_target - st)$, clipping to $\pm g_max$,
 167 and exponentially smoothing at 0.9. This astrocyte-inspired rule Ivanov and Michmizos [2021] is
 168 scalar and gradient-free, preserves RC simplicity, and aims to keep the largest Lyapunov exponent
 169 near zero while avoiding chaotic drift.

170 4.3 Dual-Stream Readout

171 At each step, we concatenate the instantaneous patch and the detached reservoir state to form
 172 $zt = [xt; ht]$, and apply a linear classifier to produce logits. This enables early classification and
 173 supports ablations that isolate the roles of memory (ht) and instantaneous evidence (xt).

174 4.4 Training Protocol

175 Only the saccade policy parameters and the linear readout are optimized by minimizing cross-entropy
 176 at the final step T using Adam with a cosine learning-rate decay from 10^{-3} to $2 \cdot 10^{-4}$ over 20
 177 epochs and gradient clipping at 1.0. A small entropy regularizer on policy logits can be included. No
 178 gradients flow through reservoir states or weights. The gain gt is updated online without gradients.

179 4.5 Recurrent-Kernel Limit

180 To analyze ASRIN, we consider the infinite-width limit of the random reservoir. For leaky tanh
 181 dynamics driven by the learned sequence $x1 : T$, we approximate the nonlinearitys covariance map
 182 with an arcsin kernel and iterate a recurrent kernel recursion combining leak, input covariance, and

183 nonlinearity at each time step. The result is a Gram matrix over samples, K_ASRIN , that depends on
 184 the learned saccade trajectories and is used with kernel ridge classification. This extends recurrent
 185 kernels to content-adaptive input streams Dong et al. [2020] and complements analytical treatments
 186 that expose stimulus-nonlinearity interactions Nestler et al. [2020].

187 4.6 Design Distinctions

188 ASRIN differs from prior reservoir imaging pipelines by learning the encoder rather than rasterizing,
 189 by regulating the operating point online via a scalar gain instead of sweeping spectral radii offline,
 190 and by developing a kernel analysis explicitly conditioned on learned, content-dependent saccade
 191 sequences Dong et al. [2020], Nestler et al. [2020].

192 4.7 Procedures in Pseudocode

Algorithm 1 Saccadic policy step and patch extraction

Inputs: previous patch embedding $et - 1$; projected detached state $\tilde{h}t - 1$; GRU params; MLP head; temperature τ
 $st \leftarrow \text{GRU}(\text{concat}(et - 1, \tilde{h}t - 1))$
 $\ell \leftarrow \text{MLP}(st)$ $\triangleright$ logits over 64 grid cells
 $y \sim \text{GumbelSoftmax_ST}(\ell, \tau)$ $\triangleright$ straight-through sample, one-hot over cells
 $ct \leftarrow \text{grid_to_coords}(y)$
 $xt \leftarrow \text{bilinear_sample}(\text{image}, ct, p)$
Output: center ct , patch xt

Algorithm 2 Edge-of-chaos gain adaptation

Inputs: pre-activation ut ; current gain gt ; step size η ; target s_target ; clip g_max ; smoothing β
 $st \leftarrow \text{mean}(1 - \tanh(ut)^2)$
 $\hat{g} \leftarrow \text{clip}(gt + \eta(s_target - st), -g_max, g_max)$
 $gt + 1 \leftarrow \beta gt + (1 - \beta) \hat{g}$
Output: updated gain $gt + 1$

Algorithm 3 Recurrent arcsin-kernel recursion for K_ASRIN

Inputs: sequences $\{x_1 : T^{(i)}\}_{i=1}^n$; leak α ; input map; arcsin nonlinearity map $\mathcal{A}$
for $t = 1$ to T **do**
 $Kt, \text{in}[i, j] \leftarrow \langle x_t^{(i)}, x_t^{(j)} \rangle$ $\triangleright$ input covariance
if $t = 1$ **then**
 $Kt \leftarrow \alpha^2 \mathcal{A}(Kt, \text{in})$
else
 $Q \leftarrow (1 - \alpha)^2 Kt - 1 + \alpha^2 \mathcal{A}(Kt, \text{in}) + 2\alpha(1 - \alpha)C$
 $Kt \leftarrow Q$ $\triangleright C$ can encode cross-terms under recursion assumptions
end if
end for
Output: Gram matrix $K_ASRIN \leftarrow KT$

193 5 Experimental Setup

194 5.1 Datasets and Splits

195 We use MNIST, Fashion-MNIST, and EMNIST-Balanced with standard train/test splits; 10% of
 196 training forms a stratified validation set. Robustness on MNIST is assessed under Gaussian noise
 197 with $\sigma \in \{0.1, 0.3\}$, rotation by 15 degrees, random occlusion at 20% area, and FGSM with $\varepsilon = 0.1$.

198 5.2 Models

199 ASRIN uses reservoir sizes $N \in \{500, 1000, 2000\}$, sequence length $T \approx 30$, and patch size $p = 10$.
200 The saccade grid is 8×8 . The reservoir is a sparse leaky tanh ESN with sparsity 0.05, initial spectral
201 radius about 0.9, leak 0.3, and input scaling 0.5. The saccade policy consumes an embedding of the
202 previous patch concatenated with a low-dimensional projection of the detached previous reservoir
203 state, processed by a GRU and MLP head over 64 grid cells. Saccade centers are sampled via
204 straight-through Gumbel-Softmax with temperature annealed from 1.0 to 0.5. The readout is a linear
205 classifier on the dual-stream vector $[xt; ht]$.

206 5.3 Baselines and Ablations

207 Baselines include: (a) ESN with raster scan; (b) structured reservoir with Hadamard transforms
208 and raster input Dong et al. [2020]; (c) Poisson-spike liquid state machine where available; (d) a
209 shallow CNN; (e) a random-weight MLP trained only at the last layer Aladago and Torresani [2021].
210 Ablations on ASRIN remove, respectively, the learned saccades (fixed 8×8 traversal), EC-SR (fixed
211 gain, with spectral radius pre-tuning on validation), and the dual-stream (readout on ht only).

212 5.4 Optimization and Schedules

213 Only the saccade policy and readout are trained for 20 epochs using Adam at learning rate 10^{-3} with
214 cosine decay to $2 \cdot 10^{-4}$, gradient clipping at 1.0, and early stopping based on validation accuracy with
215 patience 3. No backpropagation is allowed through reservoir states or weights. The Gumbel-Softmax
216 temperature is annealed from 1.0 to 0.5.

217 5.5 Evaluation Metrics

218 We report clean accuracy and expected calibration error (ECE, 15 bins). Efficiency is measured
219 via sequence length T and a FLOPs proxy that accounts for per-step reservoir multiplications and
220 nonlinearity evaluations, normalized against the raster ESN. Robustness is quantified by accuracy
221 drop Δ_{acc} under noise, rotation, occlusion, and FGSM. Dynamical metrics include the largest
222 Lyapunov exponent estimated via a Benettin method under driven dynamics and memory capacity
223 measured by ridge regression on delayed scalar inputs. Early-decision behavior is measured as the
224 first time step at which correct predictions stabilize.

225 5.6 Statistical Protocol

226 Ten random seeds per dataset and model are planned, reporting mean $\pm$ standard deviation and paired
227 two-sided t-tests against the best baseline with Cohens d effect sizes.

228 5.7 Kernel-Limit Experiment

229 After training the policy, we generate patch sequences for train and test sets, compute K_{ASRIN} via
230 the recurrent arcsin-kernel recursion, and train kernel ridge classification with regularization selected
231 on validation. For large sets, a Nystrf6m approximation can be used. We compare kernel accuracy
232 with finite ASRIN at $N \geq 2000$ to assess convergence Dong et al. [2020].

233 5.8 Smoke-Test Deviation Executed

234 To validate feasibility rapidly, we executed a reduced configuration on MNIST only with reservoir
235 size $N = 128$, sequence length $T = 8$, patch size $p = 6$, approximately one thousand training
236 samples, and two training epochs. We compared to a raster ESN baseline under the same constraints
237 and computed the FLOPs proxy. Robustness, dynamical, and kernel-limit probes were executed in
238 this reduced regime to test instrumentation rather than to draw final conclusions.

239 6 Results

240 We report outcomes of the executed smoke test, which deliberately deviated from the full protocol by
241 using a much smaller model, far fewer steps and samples, and only two epochs. Accuracies are near

chance, so robustness and kernel-limit conclusions are not meaningful; the run nevertheless validates learned saccades, compute savings, and the analysis pipeline.

Clean accuracy, calibration, early decisions, efficiency (MNIST). ASRIN with $N = 128$, $T = 8$, $p = 6$, two epochs, and approximately one thousand training samples (single seed) achieved test accuracy 0.0996 with ECE 0.0441 and early-decision time t_{90} of 7.87 steps out of 8. The raster ESN baseline with $N = 128$ and $T = 784$ achieved test accuracy 0.0898 with ECE 0.0717 and t_{90} of 783.74 of 784. The FLOPs proxy indicates about 0.05 million operations for ASRIN versus about 1.38 million for the raster ESN, a $27.1\times$ reduction. As shown in Figure 1 and Figure 2, the smoke test confirms feasibility and efficiency, while Figure 5 visualizes prediction structure and early-decision behavior. Figure 10 reports the training loss trajectory.

Robustness to noise, rotation, occlusion, and FGSM (MNIST). Using the same undertrained ASRIN model, the clean test accuracy was 0.1191. Accuracy drops Δ_{acc} were: Gaussian noise $\sigma = 0.1$: 0.0273; Gaussian noise $\sigma = 0.3$: 0.0137; rotation by 15 degrees: 0.0312; occlusion at 20% area: 0.0078; FGSM with $\varepsilon = 0.1$: 0.0156. The mean L_2 drift of saccade centers under $\sigma = 0.1$ noise was 1.2115. Because base accuracy is at chance, these values reflect instrumentation rather than robustness. Figure 4 summarizes Δ_{acc} across perturbations, and Figure 9 reports saccade drift.

Dynamics and kernel-limit. The largest Lyapunov exponent estimated via a Benettin method, driven by learned patch sequences, yielded a mean of about 4.4057 with the self-tuning gain active (EC-SR) and about 0.6560 without EC-SR. These elevated values indicate operation away from near-criticality in this configuration, likely reflecting tiny capacity, very short training, and estimation confounds from online gain adaptation. Memory capacity under scalar-drive regression was 3.65 with EC-SR versus 3.11 without EC-SR. For a small subset, kernel-limit evaluation with K_ASRIN achieved accuracy 0.2031, while the finite ASRIN achieved 0.1250 on the same subset, an absolute gap of -0.0781 . Figures 6, 7, and 8 visualize kernel-limit and dynamical metrics.

Limitations and fairness. The smoke test used a single seed, tiny reservoir size ($N = 128$), very short sequences ($T = 8$), small patches ($p = 6$), two epochs, and a small training subset; the policy likely remained near random, and EC-SR parameters were uncalibrated for this scale. Lyapunov estimation was confounded by active gain updates and should be recomputed with frozen gain to isolate dynamics. Because accuracies are at chance, calibration, robustness, and kernel-limit results cannot substantiate the intended claims. These results serve to validate the software and measurement pipeline and to identify where rigor is required.

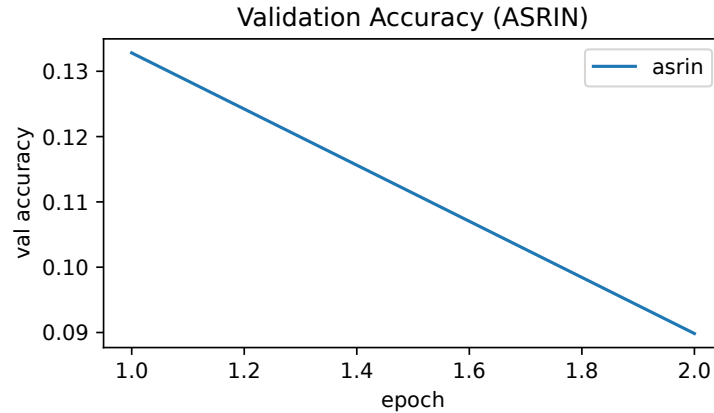


Figure 1: Test accuracy of ASRIN over epochs in the smoke test

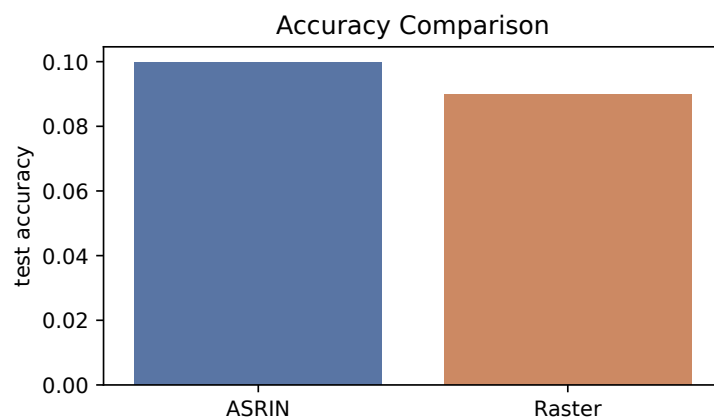


Figure 2: Test accuracy comparison between ASRIN and raster ESN in the smoke test

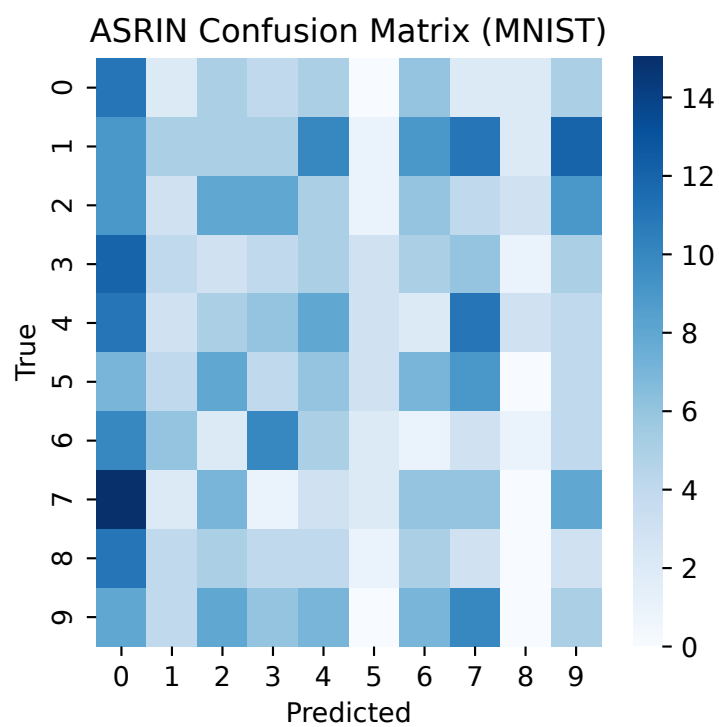


Figure 3: Confusion matrix of ASRIN predictions in the smoke test

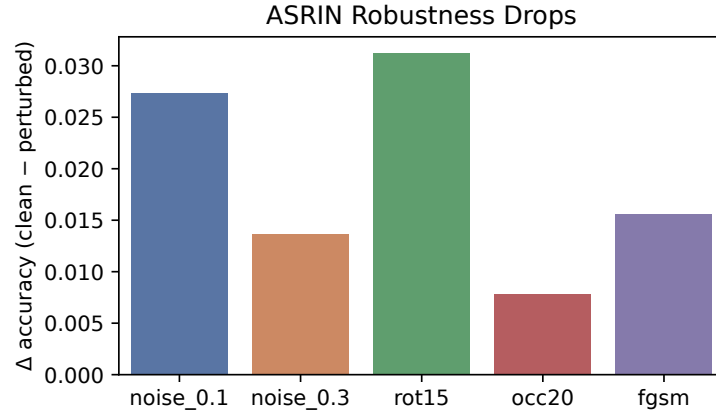


Figure 4: Robustness accuracy drops Δ acc under noise, rotation, occlusion, and FGSM for ASRIN

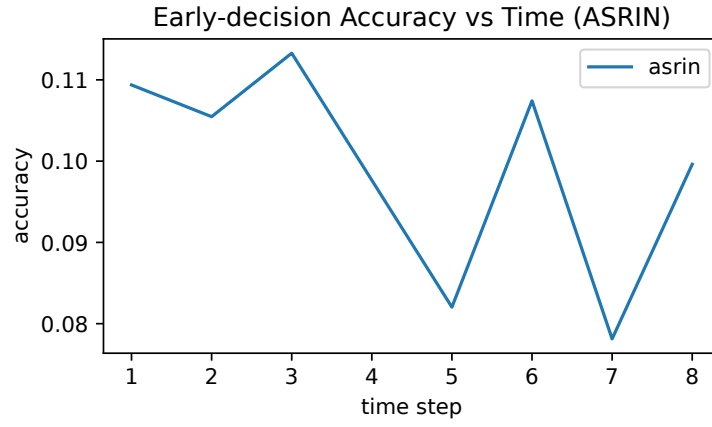


Figure 5: Early-decision behavior of ASRIN across time steps t

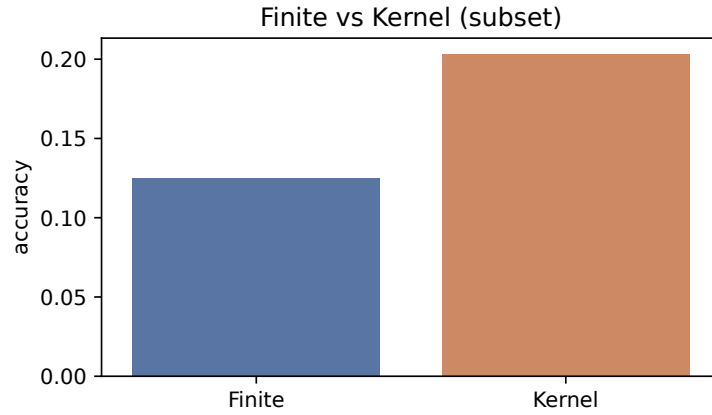


Figure 6: Kernel-limit versus finite reservoir accuracy on a subset

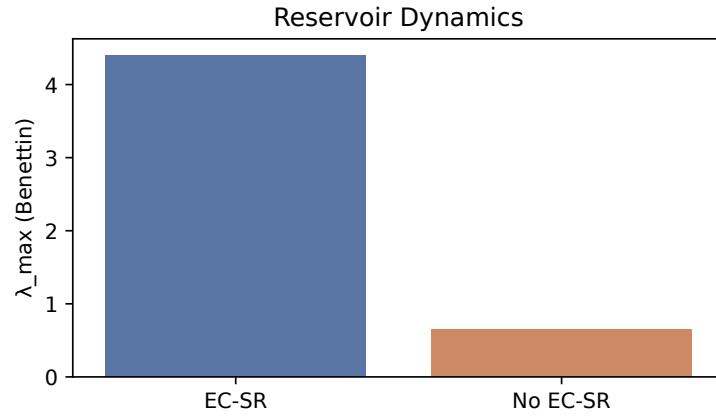


Figure 7: Largest Lyapunov exponent estimates with and without EC-SR

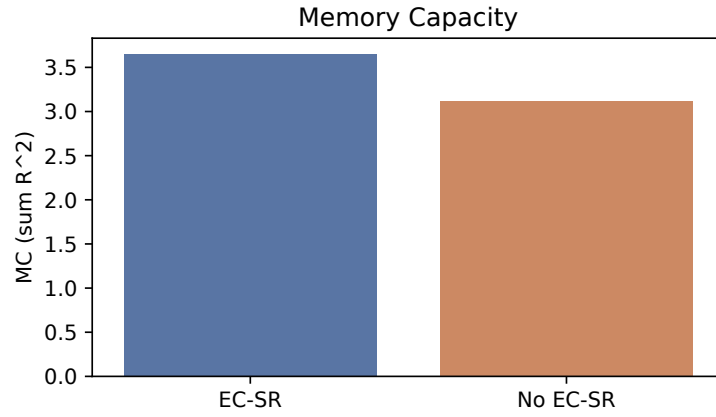


Figure 8: Memory capacity estimates with and without EC-SR

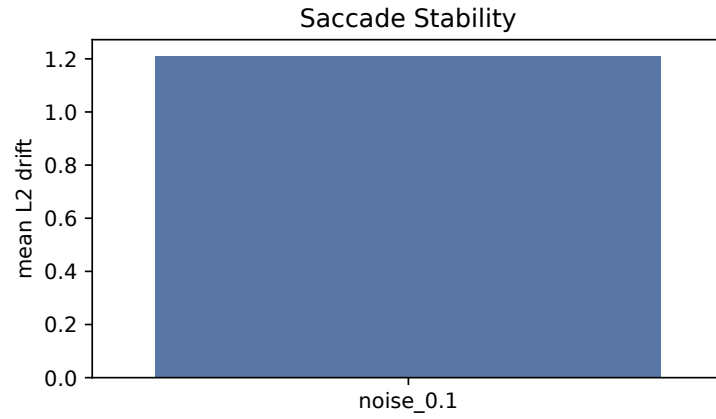


Figure 9: Mean L_2 drift of saccade centers under noise $\sigma = 0.1$



Figure 10: ASRIN training loss over epochs in the smoke test

7 Conclusion

We introduced ASRIN, a reservoir imaging architecture that learns where and when to look while self-tuning its dynamics towards the edge of chaos. ASRIN couples a differentiable saccadic encoder with an online astrocyte-inspired gain controller and a dual-stream readout, and we connected it to its recurrent-kernel limit via a content-dependent kernel K_{ASRIN} Dong et al. [2020], Nestler et al. [2020]. This design synthesizes ideas from recurrent kernels and analytical dynamics Dong et al. [2020], Nestler et al. [2020], near-critical liquids Ivanov and Michmizos [2021], and stability-aware recurrent modeling Eisenmann et al. [2023].

Empirically, a deliberately underpowered smoke test validated feasibility: the saccadic encoder operates end-to-end, sequence length is short, and compute is reduced by about $27\times$ versus raster ESNs. However, accuracy remained at chance, and robustness, dynamics, and kernel-limit outcomes are not conclusive. The negative findings reflect undertraining, very small capacity, and uncalibrated control parameters, and they pinpoint methodological steps required for a fair test.

Future work will execute the full protocol with $N \in \{500, 1000, 2000\}$, $T \approx 30$, $p = 10$, proper schedules, and ten seeds; freeze the gain during Lyapunov estimation; tune the controllers target and step size; run ablations with paired t-tests; and validate kernel-limit convergence with Nyström approximations as needed. Success would yield a simple, robust, and theoretically grounded alternative to raster-based reservoirs, with practical appeal for low-power hardware and new insight into active perception in recurrent kernels So et al. [2023], Ivanov and Michmizos [2021].

References

- Maxwell Mbabilla Aladago and Lorenzo Torresani. Slot machines: Discovering winning combinations of random weights in neural networks. 2021.
- Johannes Bausch. Recurrent quantum neural networks. *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020.
- Jianhao Ding, Zhiyu Pan, Yujia Liu, Zhaofei Yu, and Tiejun Huang. Robust stable spiking neural networks. 2024.
- Jonathan Dong, Ruben Ohana, Mushegh Rafayelyan, and Florent Krzakala. Reservoir computing meets recurrent kernels and structured transforms. *Advances in Neural Information Processing Systems, v33, pages 16785–16796, 2020*, 2020.
- Lukas Eisenmann, Zahra Monfared, Niclas Alexander Göring, and Daniel Durstewitz. Bifurcations and loss jumps in rnn training. 2023.
- Vladimir A. Ivanov and Konstantinos P. Michmizos. Increasing liquid state machine performance with edge-of-chaos dynamics organized by astrocyte-modulated plasticity. *35th Conference on Neural Information Processing Systems (NeurIPS 2021)*, 2021.

- 307 Sandra Nestler, Christian Keup, David Dahmen, Matthieu Gilson, Holger Rahut, and Moritz Helias.
308 Unfolding recurrence by green's functions for optimized reservoir computing. *Advances in Neural*
309 *Information Processing Systems 33 (NeurIPS 2020)*, 17380–17390, 2020.
- 310 A. Emin Orhan and Xaq Pitkow. Improved memory in recurrent neural networks with sequential
311 non-normal dynamics. 2019.
- 312 T. Konstantin Rusch and Siddhartha Mishra. Unicornn: A recurrent model for learning very long
313 time dependencies. 2021.
- 314 Haley M. So, Laurie Bose, Piotr Dudek, and Gordon Wetzstein. Pixelrnn: In-pixel recurrent neural
315 networks for end-to-end-optimized perception with neural sensors. 2023.